

Lecture 3: Calculus and Linear Algebra

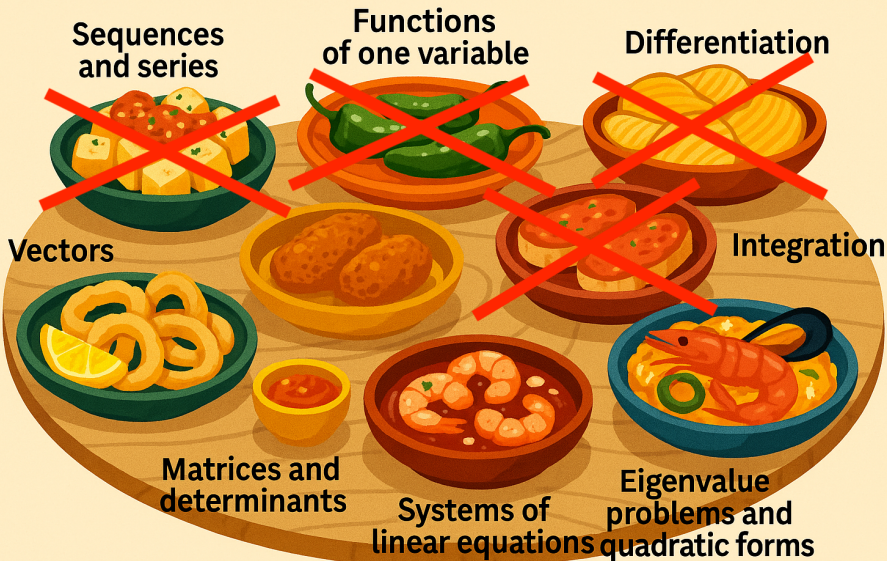
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Course content (math tapas)



Orthogonality and linear independence

Orthogonality

Definition: $\mathbf{u} \perp \mathbf{v}$ if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Angle example: For $\mathbf{u} = (2, -1, 3)$ and $\mathbf{v} = (5, -4, -1)$,

$$\langle \mathbf{u}, \mathbf{v} \rangle = 11, \quad \cos \angle(\mathbf{u}, \mathbf{v}) = \frac{11}{\sqrt{14}\sqrt{42}} \approx 0.4537,$$

so the angle is about 63° .

Geometric test: $\mathbf{u} \perp \mathbf{v}$ if and only if $\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Cauchy-Schwarz via completing the square

Assume nonzero $\mathbf{u}, \mathbf{v}$. For any real t ,

$$0 \leq \|\mathbf{u} - t\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2t \langle \mathbf{u}, \mathbf{v} \rangle + t^2 \|\mathbf{v}\|^2.$$

Complete the square in t :

$$\|\mathbf{u}\|^2 - 2t \langle \mathbf{u}, \mathbf{v} \rangle + t^2 \|\mathbf{v}\|^2 = \|\mathbf{v}\|^2 \left(t - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \right)^2 + \left(\|\mathbf{u}\|^2 - \frac{\langle \mathbf{u}, \mathbf{v} \rangle^2}{\|\mathbf{v}\|^2} \right).$$

Since the left side is ≥ 0 for all t , the second term must be ≥ 0 :

$$\langle \mathbf{u}, \mathbf{v} \rangle^2 \leq \|\mathbf{u}\|^2 \|\mathbf{v}\|^2.$$

Equality condition: equality holds if and only if $\|\mathbf{u} - t\mathbf{v}\|^2 = 0$ for $t = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}$, i.e., $\mathbf{u} = t\mathbf{v}$ (collinear vectors).



Linear dependence, independence, and bases

Linear combination: $\mathbf{u} = \sum_{i=1}^m \lambda_i \mathbf{v}_i$.

Linear independence: $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is linearly independent if

$$\sum_{i=1}^m \lambda_i \mathbf{v}_i = \mathbf{0} \quad \Rightarrow \quad \lambda_1 = \dots = \lambda_m = 0.$$

Basis: Any n linearly independent vectors in $\mathbb{R}^n$ form a basis. Then every $\mathbf{u} \in \mathbb{R}^n$ has a **unique** representation $\mathbf{u} = \sum_{i=1}^n \lambda_i \mathbf{v}_i$.

Standard basis: $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$.

Subspaces, span, and dimension

For $E = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$, the **span** is

$$\text{span}(E) = \left\{ \sum_{i=1}^k \lambda_i \mathbf{v}_i : \lambda_i \in \mathbb{R} \right\}.$$

A **subspace** $V \subset \mathbb{R}^n$ is any span. A **basis of V** is a linearly independent set that spans V . The **dimension** of V is the size of any basis.

Exercise: Basis of $V = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$.

Orthogonal complement and projection

Orthogonal complement

For a subspace $V \subset \mathbb{R}^n$,

$$V^\perp = \{\mathbf{x} : \langle \mathbf{x}, \mathbf{v} \rangle = 0 \ \forall \mathbf{v} \in V\}.$$

Projection theorem

For any $\mathbf{y} \in \mathbb{R}^n$ and subspace V , there is a unique $\hat{\mathbf{y}} \in V$ with $\mathbf{y} - \hat{\mathbf{y}} \in V^\perp$. We call $\hat{\mathbf{y}}$ the orthogonal projection of $\mathbf{y}$ onto V .

Least squares view: With design matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and response $\mathbf{y}$, the LS fit $\hat{\mathbf{y}}$ is the projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$.

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$



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Chapter 6: Matrices and determinants

A **matrix** $A \in \mathbb{R}^{m \times n}$ is a table with m rows, n columns. The (i, j) entry is a_{ij} .

We recall basic facts about matrices.

Why care: matrices express linear maps, data tables, network flows, input-output models, regressions, transformations, and more.

Read Werner-Sotskov Ch. 7; Simon-Blume Chs. 8-9.

Exercises 7.6, 7.9(b,c,d), 7.12, 7.14(a), 7.16, 7.18

Matrices

A table of numbers with m rows and n columns: $A \in \mathbb{R}^{m \times n}$. The (i, j) -th entry is denoted by a_{ij} .

Special matrices:

- zero matrix: $\mathbf{0}_{m \times n} \in \mathbb{R}^{m \times n}$
- identity matrix: $\mathbb{I}_n \in \mathbb{R}^{n \times n}$.
- **transposition**: $A^T \in \mathbb{R}^{n \times m}$, $(A^T)_{ij} = a_{ji}$.
- square matrix: $m = n$.
- symmetric matrix: square matrix such that $A = A^T$.
- diagonal matrix: $a_{ij} = 0$ if $i \neq j$.
- lower (upper) triangular: $a_{ij} = 0$ if $i < j$ ($i > j$).
- $\mathbf{v} \in \mathbb{R}^n$ is treated as $n \times 1$ matrix, $\mathbb{R}^n \simeq \mathbb{R}^{n \times 1}$.

Basic matrix operations

Addition and scalar multiplication are entrywise:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (\lambda A)_{ij} = \lambda a_{ij}.$$

They satisfy the usual commutative, associative, and distributive laws.

Matrix product: If $A \in \mathbb{R}^{m \times p}$ and $B \in \mathbb{R}^{p \times n}$, then $C = AB \in \mathbb{R}^{m \times n}$ with

$$c_{ij} = \sum_{k=1}^p a_{ik} b_{kj}.$$

Example:

$$\begin{pmatrix} 2 & 3 & 4 & 1 \\ 7 & -1 & 0 & 4 \end{pmatrix} \begin{pmatrix} 2 & 7 \\ 3 & -1 \\ 4 & 0 \\ 1 & 4 \end{pmatrix} = \begin{pmatrix} 30 & 15 \\ 15 & 66 \end{pmatrix}.$$

Algebraic properties

Facts:

1. $(AB)C = A(BC)$
2. $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$
3. Generally **not** commutative: $AB \neq BA$
4. $AI_n = I_m A = A$ (dimensions must match)
5. $(A^T)^T = A$, $(A + B)^T = A^T + B^T$, $(\lambda A)^T = \lambda A^T$, $(AB)^T = B^T A^T$

Remark: AA^T is always symmetric.

Matrix times vector

We treat $\mathbb{R}^n$ as column vectors $\mathbb{R}^{n \times 1}$. If $A \in \mathbb{R}^{m \times n}$ and $\mathbf{x} \in \mathbb{R}^n$ then $A\mathbf{x} \in \mathbb{R}^m$ is a linear combination of the columns of A with coefficients x_i :

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n.$$

e.g. **Another look at the LS method:** $\mathbf{X} \in \mathbb{R}^{n \times d}$, $\mathbf{y} \in \mathbb{R}^n$

$$\text{minimize}_{\beta \in \mathbb{R}^d} \quad \|\mathbf{y} - \mathbf{X}\beta\|^2$$

Finds the closest point to $\mathbf{y}$ in the space spanned by the columns of $\mathbf{X}$.

Orthogonal projection

Theorem: Given a vector $\mathbf{y} \in \mathbb{R}^n$ and a subspace $V \subset \mathbb{R}^n$ there exists a unique $\hat{\mathbf{y}} \in V$ such that $\mathbf{y} - \hat{\mathbf{y}} \in V^\perp$. Let $\mathbf{x}_1, \dots, \mathbf{x}_d$ be a basis of $V \subset \mathbb{R}^n$.

$\mathbf{X} \in \mathbb{R}^{n \times d}$ with columns $\mathbf{x}_1, \dots, \mathbf{x}_d$.

$\hat{\mathbf{y}} \in V$ means $\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\lambda}$ for some $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_d)$.

$\mathbf{y} - \hat{\mathbf{y}} \in V^\perp$ means $\mathbf{X}^\top(\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{0}$.

The unique solution: $\boldsymbol{\lambda} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ giving

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Matrix inverse

A square matrix A is **invertible** if there exists a matrix A^{-1} such that

$$AA^{-1} = A^{-1}A = I.$$

We want to show:

$$A \in \mathbb{R}^{n \times n} \text{ is invertible} \iff A\mathbf{x} = \mathbf{0}_n \text{ only for } \mathbf{x} = \mathbf{0}_n.$$

Two observations:

1. $A\mathbf{x} = \mathbf{0}$ only for $\mathbf{x} = \mathbf{0}$ iff columns of A are lin. independent.
2. n independent vectors in $\mathbb{R}^n$ form a basis and so $\forall i = 1, \dots, n \exists \mathbf{b}_i$ such that $A\mathbf{b}_i = \mathbf{e}_i$.
3. This gives $B \in \mathbb{R}^{n \times n}$ such that $AB = \mathbb{I}_n$.

This is enough to show that the matrix $\mathbf{X}^\top \mathbf{X}$ on slide 14 is invertible.

Note: If $(\mathbf{X}^\top \mathbf{X})\mathbf{x} = \mathbf{0}$ then $\mathbf{x}^\top (\mathbf{X}^\top \mathbf{X})\mathbf{x} = 0$ but this only possible if $\mathbf{x} = \mathbf{0}$.

Important spaces and rank

For $A \in \mathbb{R}^{m \times n}$:

- **Column space** (image) $\text{Im}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} \subset \mathbb{R}^m$
- **Row space** $\text{Im}(A^T) \subset \mathbb{R}^n$
- **Kernel** (null space) $\ker(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\} \subset \mathbb{R}^n$
- **Rank** $\text{rank}(A) = \dim \text{Im}(A) = \dim \text{Im}(A^T)$

Orthogonality: Row space is orthogonal to $\ker(A)$.

Rank-nullity: $\text{rank}(A) + \dim \ker(A) = n$.

Try: $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$.

1	14	14	4
11	7	6	9
8	10	10	5
13	2	3	15



Orthogonal matrices

An $n \times n$ matrix A is called **orthogonal** if $AA^\top = I_n$.

Remark: If A is orthogonal, its row vectors (and also its column vectors) are pairwise orthogonal unit vectors.

Proof: Let $\mathbf{r}_i$ and $\mathbf{r}_j$ be the i -th and j -th rows of A . The (i, j) entry of $AA^\top$ is the scalar product $\mathbf{r}_i^\top \mathbf{r}_j$. If $AA^\top = I$, then $\mathbf{r}_i^\top \mathbf{r}_j = 0$ for $i \neq j$ (orthogonality) and $\mathbf{r}_i^\top \mathbf{r}_i = \|\mathbf{r}_i\|^2 = 1$ (unit length). The same holds for columns using $A^\top A = I_n$.

Example:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Singular value decomposition (SVD)

Singular value decomposition

For every $A \in \mathbb{R}^{m \times n}$ there exist orthogonal matrices $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ and a “diagonal” matrix $\Lambda \in \mathbb{R}^{m \times n}$ such that

$$A = U\Lambda V^T.$$

If $m \leq n$ then Λ has m “diagonal” entries.

We can assume $\Lambda_{11} \geq \dots \geq \Lambda_{mm} \geq 0$.

These diagonal entries are called **singular values** of A .

Spectral decomposition

Singular value decomposition

Suppose $A \in \mathbb{R}^{n \times n}$ is symmetric. Then there exist orthogonal $U \in \mathbb{R}^{n \times n}$ and a diagonal Λ such that

$$A = U\Lambda U^{\top}.$$

Denote $(\lambda_1, \dots, \lambda_n) = \text{diag}(\Lambda)$.

We take $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m$.

Elementary matrix operations

Column or row operations:

1. **Swap** two columns (or rows)
2. **Scale** a column (or row) by $\lambda \neq 0$
3. **Add** a multiple of one column (or row) to another

Each is implemented by multiplying by a suitable elementary matrix on the right (for column ops) or left (for row ops). Useful for Gaussian elimination and determinant computation.

To see what is happening, let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and multiply from the right by $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} \lambda & 0 \\ 0 & 1 \end{bmatrix}$, or $\begin{bmatrix} 1 & 0 \\ \lambda & 1 \end{bmatrix}$.

Determinants: definition

For $A \in \mathbb{R}^{n \times n}$, let A_{ij} be the submatrix with row i and column j removed. Define recursively

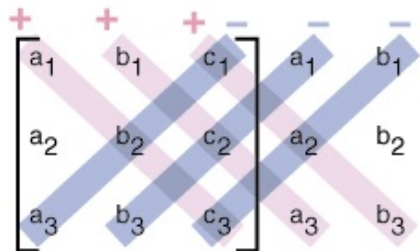
$$\det(A) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}), \quad \det([a_{11}]) = a_{11}.$$

For $n = 2$: $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$

For $n = 3$:

$$\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}.$$

A memory aid for $n = 3$



$$\det A = (a_1 b_2 c_3 + b_1 c_2 a_3 + c_1 a_2 b_3) - (a_3 b_1 c_2 + b_3 c_1 a_2 + c_3 a_2 b_1)$$

Cofactor expansion and properties

Cofactor expansion: expand by any row i or any column j :

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}).$$

Properties:

1. $\det(A) = \det(A^T)$
2. If A is triangular, $\det(A)$ is the product of diagonal entries
3. $\det(AB) = \det(A) \det(B)$
4. Swapping two rows (or columns) flips the sign of $\det$
5. Scaling a row (or column) by λ scales $\det$ by λ
6. Adding a multiple of one row to another leaves $\det$ unchanged
7. $\det(A) = 0$ if and only if rows (or columns) are linearly dependent

Determinants by elimination

Use row operations (keeping track of determinant changes) to reach triangular form.

$$\begin{vmatrix} 1 & 2 & 0 \\ 3 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 0 \\ 0 & -6 & 1 \\ 1 & 2 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 0 \\ 0 & -6 & 1 \\ 0 & 0 & 3 \end{vmatrix} = (-6) \cdot 3 = -18.$$

First: $r_2 \leftarrow r_2 - 3r_1$. Then: $r_3 \leftarrow r_3 - r_1$.

Geometric meaning: $|\det(A)|$ is the area/volume scaling of the linear map $x \mapsto Ax$ (and its sign encodes orientation).

Linear systems and Cramer's rule

A system $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{R}^{n \times n}$, unknown $\mathbf{x} \in \mathbb{R}^n$, and data $\mathbf{b}$. If A is **nonsingular** ($\det A \neq 0$), the solution is unique.

Cramer's rule: Let $A_j(\mathbf{b})$ be A with column j replaced by $\mathbf{b}$. Then

$$x_j = \frac{\det A_j(\mathbf{b})}{\det A}, \quad j = 1, \dots, n.$$

Note: Great for theory, not used for large-scale computation.

Example:

$$A = \begin{pmatrix} 2 & 3 & 5 \\ 1 & 0 & 2 \\ -1 & -4 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 7 \\ 4 \end{pmatrix} \Rightarrow \mathbf{x} = \left(\frac{75}{8}, -\frac{63}{16}, -\frac{19}{16} \right).$$

Linear mappings

A mapping $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **linear** if

$$A(\mathbf{x}_1 + \mathbf{x}_2) = A(\mathbf{x}_1) + A(\mathbf{x}_2), \quad A(\lambda \mathbf{x}) = \lambda A(\mathbf{x}).$$

Then there exists an $m \times n$ matrix (also denoted A) with $A(\mathbf{x}) = A\mathbf{x}$.

Columns as images: $A(\mathbf{e}_i) = \mathbf{a}_i$ (the i -th column), and $A(\mathbf{x}) = \sum_i x_i A(\mathbf{e}_i)$.

Examples: scalings, rotations, reflections, projections, feature maps in ML, Leontief input-output in economics.

Two simple linear maps

1. Reflection across the y -axis:

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ y \end{pmatrix}.$$

2. Rotation by 45° counterclockwise:

$$A = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Check the images of $\mathbf{e}_1$ and $\mathbf{e}_2$ to see the action.

Inverse matrix and properties

A square A is invertible if A^{-1} exists with $AA^{-1} = A^{-1}A = I_n$.

Properties:

1. $(A^{-1})^{-1} = A$
2. $(AB)^{-1} = B^{-1}A^{-1}$
3. $(A^T)^{-1} = (A^{-1})^T$
4. $(\lambda A)^{-1} = \lambda^{-1}A^{-1}$ for $\lambda \neq 0$
5. $\det(A^{-1}) = 1/\det(A)$

Solve $A\mathbf{x} = \mathbf{b}$ by $\mathbf{x} = A^{-1}\mathbf{b}$ when A is invertible.

Computing inverses

Cofactor formula: If A is nonsingular,

$$A^{-1} = \frac{1}{\det A} C(A)^T, \quad C(A)_{ij} = (-1)^{i+j} \det(A_{ij}).$$

For 2×2 :

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

For larger n , numerical methods use elimination (LU), not cofactors.

Example: Input-output model

Let a_{ij} be units of good i needed to produce 1 unit of good j . Put $A = (a_{ij})$. Let $\mathbf{x}$ be total output and $\mathbf{y}$ the vector of final demand.

Accounting identity: output = internal demand + final demand

$$\mathbf{x} = A\mathbf{x} + \mathbf{y} \quad \Leftrightarrow \quad (I_n - A)\mathbf{x} = \mathbf{y} \quad \Rightarrow \quad \mathbf{x} = (I_n - A)^{-1}\mathbf{y}$$

provided $I_n - A$ is invertible.

Interpretation: $(I - A)^{-1} = I + A + A^2 + \dots$ accumulates direct, indirect, and higher-order input needs when it converges.

A triangular example

Theorem: If A is strictly upper triangular (zeros on and below diagonal), then $A^n = 0$ and

$$(I_n - A)^{-1} = I_n + A + A^2 + \cdots + A^{n-1}.$$

Check $(I - A)(I + A + \cdots + A^{n-1}) = I - A^n = I$.

Example:

$$A = \begin{pmatrix} 0 & 3 & 5 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 1 \\ 7 \\ 4 \end{pmatrix} \Rightarrow \mathbf{x} = (66, 15, 4).$$

Chapter 7: Systems of Linear Equations

Many problems in economics and data science can be modeled as a **system of linear equations**.

In this chapter we recall core facts and **general solution procedures**.

Read Chapter 8 of Werner–Sotskov; Chapter 7 of Simon–Blume.

Exercises 8.5, 8.6, 8.8 (Werner–Sotskov).

Systems of linear equations

Let A be an $m \times n$ matrix, $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ unknown, and $\mathbf{b} \in \mathbb{R}^m$ given. The system

$$A\mathbf{x} = \mathbf{b}$$

has the **vector form**

$$\sum_{j=1}^n x_j \mathbf{a}_j = \mathbf{b},$$

where $\mathbf{a}_1, \dots, \mathbf{a}_n$ are the columns of A .

If $\mathbf{b} = \mathbf{0}$ the system is **homogeneous**; otherwise it is **inhomogeneous**.

An inhomogeneous system can be **inconsistent** (no solution) or **consistent** (at least one solution).

A homogeneous system always has $\mathbf{x} = \mathbf{0}$ as a solution.

Motivating example: ranking the web (PageRank)

Modern search engines rank pages by solving a linear system. Build a row-stochastic transition matrix P where P_{ij} is the probability to move from page i to page j . The long-run visit probabilities $\mathbf{x}$ solve

$$(I - \alpha P^\top) \mathbf{x} = (1 - \alpha) \mathbf{v}, \quad 0 < \alpha < 1,$$

with $\mathbf{v}$ a teleport vector (often uniform).

Mini-web example: pages A, B, C, D with links $A \rightarrow \{B, C\}$, $B \rightarrow C$, $C \rightarrow A$, $D \rightarrow \{A, C\}$. Then

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{pmatrix}, \quad \alpha = 0.85, \quad \mathbf{v} = \frac{1}{4}(1, 1, 1, 1)^\top.$$

Solving gives

$$\mathbf{x} \approx (0.380, 0.199, 0.384, 0.038),$$

so C and A rank highest.

Node D is rarely visited because nobody links to it (only teleportation contributes).

Motivating example: deblurring a photo

Phone photos often suffer from motion blur. Blurring is (approximately) linear:

$$b = Ax + \text{noise},$$

where x is the unknown sharp image (vectorized), b is the observed blurred image, and A encodes the blur kernel (convolution).

- **Ideal case:** noise-free, known $A \Rightarrow$ solve the linear system $Ax = b$.
- **Realistic case:** noisy or overdetermined $\Rightarrow$ solve a least-squares problem

$$\min_x \|Ax - b\|^2 + \lambda \|x\|^2,$$

which leads to the normal equations $(A^\top A + \lambda I_n)x = A^\top b$.

Same linear algebra ideas: vector form, consistency, rank, and algorithms to solve large systems.

Rank and existence/uniqueness

Theorem (Column rank = Row rank)

The maximum number of linearly independent columns of A equals the maximum number of linearly independent rows. This common number is the **rank** of A , denoted $\text{rank}(A)$.

Fact: $\text{rank}(A)$ equals the largest order of a **minor** (determinant of a square submatrix) that is nonzero.

Example:

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 4 & 6 & 2 \\ 3 & 2 & 4 \end{pmatrix}$$

Here $\det(A) = 0$, so $\text{rank}(A) \leq 2$. Since $\det(A_{33}) = \det \begin{pmatrix} 1 & 2 \\ 4 & 6 \end{pmatrix} = -2 \neq 0$, we get $\text{rank}(A) = 2$.

Note $2\mathbf{a}_1 - \mathbf{a}_2 = \mathbf{a}_3$, but no single column is a multiple of another. Any two of the three columns form a basis of the column space.

Computing the rank with Gaussian elimination

Elementary row or column operations do not change $\text{rank}(A)$.

$$\begin{pmatrix} 1 & 2 & 0 \\ 4 & 6 & 2 \\ 3 & 2 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 0 \\ 0 & -2 & 2 \\ 0 & -4 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

The **pivot** in each nonzero row is the first nonzero entry from the left.

There are 2 pivots, hence $\text{rank}(A) = 2$.

To find a linear relation among columns, solve the homogeneous system for $(\lambda_1, \lambda_2, \lambda_3)$:

$$\lambda_1 + 2\lambda_2 = 0, \quad -\lambda_2 + \lambda_3 = 0.$$

Thus $(\lambda_1, \lambda_2, \lambda_3) = (-2\lambda_3, \lambda_3, \lambda_3)$ and, e.g., for $\lambda_3 = 1$,

$$-2\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 = \mathbf{0}.$$

Augmented matrix; consistency

The **augmented matrix** of the linear system $A\mathbf{x} = \mathbf{b}$ is $A_{\mathbf{b}} = [A \ \mathbf{b}] \in \mathbb{R}^{m \times (n+1)}$.

Remark: Either $\text{rank}(A_{\mathbf{b}}) = \text{rank}(A)$ or $\text{rank}(A_{\mathbf{b}}) = \text{rank}(A) + 1$.

Theorem (Rouch–Capelli): $A\mathbf{x} = \mathbf{b}$ is **consistent** $\iff \text{rank}(A_{\mathbf{b}}) = \text{rank}(A)$
(equivalently $\mathbf{b} \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$).

Moreover:

1. If $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = n$, the solution is **unique**. **Q:** What is it?
2. If $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) < n$, there are **infinitely many** solutions; choose $n - \text{rank}(A)$ variables **free**.

Triangularization (Gaussian elimination)

Theorem

Applying elementary row operations to $A_{\mathbf{b}}$ transforms any system into an equivalent **upper-triangular** one.

This follows essentially from the fact that $A\mathbf{x} = \mathbf{b} \Leftrightarrow A_{\mathbf{b}} \begin{bmatrix} \mathbf{x} \\ -1 \end{bmatrix} = \mathbf{0} \Leftrightarrow CA_{\mathbf{b}} \begin{bmatrix} \mathbf{x} \\ -1 \end{bmatrix} = \mathbf{0}$ for any invertible C

$$\begin{array}{rcl} x_1 + x_2 + x_3 = 3 & & x_1 + x_2 + x_3 = 3 \\ x_1 - x_2 + 2x_3 = 2 & \iff & -2x_2 + x_3 = -1 \\ 4x_1 + 6x_2 - x_3 = 9 & & -4x_3 = -4 \end{array}$$

Row ops: $r_2 \leftarrow r_2 - r_1$, $r_3 \leftarrow r_3 - 4r_1$, $r_3 \leftarrow r_3 + r_2$.

We have $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = 3 \Rightarrow$ unique solution $\mathbf{x} = (1, 1, 1)$.

A homogeneous example

Consider

$$\begin{aligned}a + b + c + d + e + f &= 0, \\2a + 2b + 2c + 2d - e - f &= 0, \\3a + 3b - c - d - e - f &= 0.\end{aligned}$$

Row-reducing the coefficient matrix gives

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix},$$

so there are 3 pivots, $\text{rank} = 3$, and the solution space has dimension $6 - \text{rank} = 3$:

$$(a, -a, c, -c, e, -e).$$

A basis is $\{(1, -1, 0, 0, 0, 0), (0, 0, 1, -1, 0, 0), (0, 0, 0, 0, 1, -1)\}$.

General solution structure

Theorem: Every solution of $A\mathbf{x} = \mathbf{b}$ can be written as

$$\mathbf{x} = \mathbf{x}_h + \mathbf{x}_p,$$

where $\mathbf{x}_h$ is the **general solution** of $A\mathbf{x} = \mathbf{0}$ and $\mathbf{x}_p$ is any **particular solution** of $A\mathbf{x} = \mathbf{b}$.

Corollary: If $A\mathbf{x} = \mathbf{b}$ is consistent, it has as many solutions as $A\mathbf{x} = \mathbf{0}$.

Example: Three identical equations

$$x_1 + x_2 + x_3 = 1$$

yield $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = 1$. Choose two free variables:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \lambda_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \lambda_1, \lambda_2 \in \mathbb{R}.$$

Two quick examples

1. For $A = \begin{pmatrix} 1 & 2 & 0 \\ 4 & 6 & 2 \\ 3 & 2 & 4 \end{pmatrix}$ we found $\text{rank}(A) = 2$. Thus
 $\dim \ker(A) = 3 - 2 = 1$ and

$$\ker(A) = \{\lambda(-2, 1, 1) : \lambda \in \mathbb{R}\}.$$

2. Consider

$$\begin{aligned}x_1 + x_2 + x_3 &= -1, \\x_1 + 2x_2 + 4x_3 &= 2.\end{aligned}$$

Row reduction gives $x_1 + x_2 + x_3 = -1$, $x_2 + 3x_3 = 3$, so
 $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = 2$. One free variable:

$$\{\lambda(2, -3, 1) + (-4, 3, 0) : \lambda \in \mathbb{R}\}.$$

Finding a basis for a span

Let

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 \\ 3 \\ -2 \end{pmatrix}.$$

Form the 3×3 matrix with these as **columns**:

$$\begin{pmatrix} 2 & -1 & 0 \\ 3 & 0 & -1 \\ 0 & 3 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1/3 \\ 0 & 1 & -2/3 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus $\text{rank} = 2$ and $3\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$. A convenient basis of $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ -1/3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -2/3 \end{pmatrix} \right\}$$

(or, equivalently, two of the original vectors that are independent).

Standing on the shoulders of giants



"If I have seen further, it is by standing on the shoulders of giants." **Isaac Newton (1675)**.

Some of the giants of linear/matrix algebra:

- Gauss (Gaussian elimination, least squares, determinants)
- Cauchy (CauchySchwarz, eigenvalues, determinants)
- Laplace (Laplace expansion of determinants)
- Cayley (CayleyHamilton theorem, matrix theory)
- Sylvester (matrix rank, invariant theory)
- Jordan (Jordan canonical form)
- Frobenius (matrix factorizations, linear algebra foundations)

(pictured here as the base of a *castell*)

Chapter 8: Eigenvalues and Quadratic Forms

Eigenvalues/eigenvectors appear in growth models, Markov chains, PCA, and stability of dynamical systems.

Quadratic forms drive optimization via second-order conditions.

Read Chapter 10 of Werner–Sotskov (optional: Simon–Blume, Ch. 7).

Exercises 10.2; 10.5 A,B,C; 10.6(a,b).

Eigenvalues and eigenvectors: migration example

Consider a population split between two regions: city (x_t^C) and countryside (x_t^R) at time t . Suppose each year a fraction of people move between the two regions:

$$\begin{pmatrix} x_{t+1}^C \\ x_{t+1}^R \end{pmatrix} = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix} \begin{pmatrix} x_t^C \\ x_t^R \end{pmatrix} = A \begin{pmatrix} x_t^C \\ x_t^R \end{pmatrix}.$$

Here, 0.8 means that 80% of city dwellers stay in the city, while 0.4 means that 40% of rural dwellers move to the city in the next period.

If the **urban/rural ratio stabilizes**, the next state must be a constant multiple of the current state:

$$\begin{pmatrix} x_{t+1}^C \\ x_{t+1}^R \end{pmatrix} = \lambda \begin{pmatrix} x_t^C \\ x_t^R \end{pmatrix}.$$

The long-run stable distribution of the population is given by an **eigenvector** of A , and λ is the associated **eigenvalue** describing the growth rate.

Eigenvalues and eigenvectors: definitions

For $A \in \mathbb{R}^{n \times n}$, $\lambda \in \mathbb{R}$ is an **eigenvalue** if there exists $\mathbf{x} \neq \mathbf{0}$ with

$$A\mathbf{x} = \lambda\mathbf{x} \iff (A - \lambda I_n)\mathbf{x} = \mathbf{0}.$$

Then $\mathbf{x}$ is an **eigenvector** for λ .

Characterization:

$$\lambda \text{ eigenvalue} \iff \text{rank}(A - \lambda I_n) < n \iff \det(A - \lambda I_n) = 0.$$

The **characteristic polynomial** $P(\lambda) = \det(A - \lambda I_n)$ has degree n ; the real roots (at most n) are the eigenvalues.

Procedure:

1. Solve $\det(A - \lambda I_n) = 0$ to get eigenvalues.
2. For each λ , solve $(A - \lambda I_n)\mathbf{x} = \mathbf{0}$ for eigenvectors.

Cohort-mix example (continued)

With $A = \begin{pmatrix} 0.8 & 0.4 \\ 0.3 & 0.9 \end{pmatrix}$,

$$(0.8 - \lambda)(0.9 - \lambda) - 0.12 = 0 \iff \lambda^2 - 1.7\lambda + 0.6 = 0.$$

The solutions are $\lambda_1 = 1.2$ and $\lambda_2 = 0.5$.

For $\lambda_1 = 1.2$, eigenvectors are multiples of $(1, 1)$. For $\lambda_2 = 0.5$, eigenvectors are multiples of $(1, -\frac{3}{4})$.

For a population-growth interpretation we keep the dominant eigenpair: equal cohort proportions grow by 20% per step.

Basic spectral facts

If $A \in \mathbb{R}^{n \times n}$,

1. Eigenvectors associated with distinct eigenvalues are **linearly independent**.
2. If A is **symmetric**, then all eigenvalues are real and eigenvectors for distinct eigenvalues are **orthogonal** (Spectral Theorem).

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

Then $P(\lambda) = \lambda^2 - 2\lambda - 3$, so $\lambda_1 = 3$, $\lambda_2 = -1$ with eigenvectors along $(1, 1)$ and $(-1, 1)$, respectively.

Take the spectral decomposition $A = U\Lambda U^T$. Show columns of U are the eigenvectors.

Repeated eigenvalues

Consider

$$A = \begin{pmatrix} 0 & -1 & 1 \\ -7 & 0 & 5 \\ -5 & -2 & 5 \end{pmatrix}.$$

The characteristic polynomial is

$$p_A(\lambda) = \lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda - 2)^2(\lambda - 1).$$

Hence the eigenvalues are $\lambda = 2$ (algebraic multiplicity 2) and $\lambda = 1$ (algebraic multiplicity 1).

Eigenspaces:

$$E_2 = \ker(A - 2I) = \text{span}\{(1, -1, 1)\}, \quad E_1 = \ker(A - I) = \text{span}\{(2, 1, 3)\}.$$

Since $\dim E_2 = 1 < 2$, A is defective (not diagonalizable).

Application: Principal Component Analysis (PCA)

We observe n data points $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$.

Step 1 (Centering): Arrange the data in a matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ with rows $\mathbf{x}_1, \dots, \mathbf{x}_n$. Define the centering matrix $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ and then the **centered data** is $H\mathbf{X}$.

Step 2 (Sample covariance):

$$S = \frac{1}{n}(H\mathbf{X})^\top(H\mathbf{X}).$$

Step 3 (Spectral decomposition):

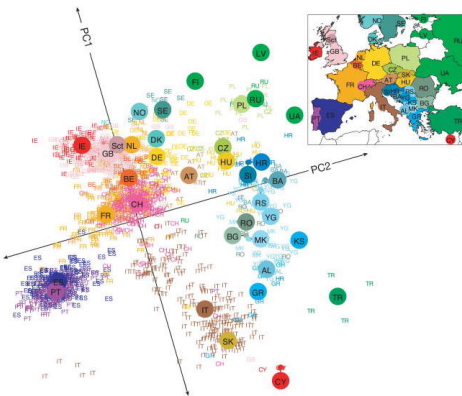
$$S = U\Lambda U^\top \quad \text{and define } \mathbf{Y} = H\mathbf{X}U \quad (\text{rotate the centered data}).$$

The sample covariance of $\mathbf{Y}$ is $\frac{1}{n}\mathbf{Y}^\top\mathbf{Y} = U^\top S U = \Lambda$ (diagonal!).

The columns of U are **principal directions** and diagonal entries of Λ are the **explained variances**.

Sorting eigenvalues in decreasing order ranks components by variance captured.

PCA: why it matters



Dimensionality reduction: compress high-dimensional data into a few directions that preserve most variability.

Noise filtering: small eigenvalues $\rightarrow$ directions with little signal.

Visualization: projecting onto first 2 PCs often reveals hidden structure.

Nice example: genetic variation in Europeans plotting individuals on the first two PCs of SNP data *recovers the geographic map of Europe!*

Numerical toy example

Covariance matrix:

$$S = \begin{pmatrix} 72.9 & 87.5 & 6.875 \\ 87.5 & 108.3 & 8.75 \\ 6.875 & 8.75 & 0.73 \end{pmatrix}.$$

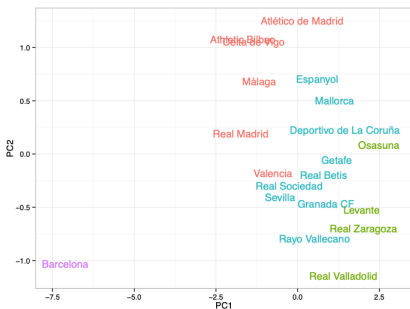
Spectral decomposition:

$$S = U\Lambda U^{\top}, \quad \Lambda = \text{diag}(180, 1.38, 0.005).$$

⇒ The first component explains > 99% of the variance; the last can be safely ignored.

PCA & Unique football styles

“Mille viae ducunt Barcinonem” (A thousand roads lead to Barcelona)



“Searching for a Unique Style in Soccer” (Gyarmati, Kwak & Rodriguez, 2014), studies sequential pass patterns in a team’s passing.

They defined “flow motifs” as statically significant pass subsequences (e.g., $A \rightarrow B \rightarrow C \rightarrow A$).

Applied motif analysis to compare styles across teams and leagues. The data are projected to the first two principal components.

Quadratic forms

A **quadratic form** is $Q(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x} = \sum_{i,j=1}^n a_{ij} x_i x_j$ with $A \in \mathbb{R}^{n \times n}$ symmetric.

Remark: Since the coefficient of $x_i x_j$ is $a_{ij} + a_{ji}$, every quadratic form has a unique symmetric representative (replace A by $\frac{1}{2}(A + A^\top)$).

Example: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $\begin{pmatrix} 1 & 5/2 \\ 5/2 & 4 \end{pmatrix}$ define the same

$$Q(x, y) = x^2 + 5xy + 4y^2.$$

Signs of quadratic forms

For symmetric A :

1. Positive definite: $\mathbf{x}^\top A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.
2. Negative definite: $\mathbf{x}^\top A \mathbf{x} < 0$ for all $\mathbf{x} \neq \mathbf{0}$.
3. Positive semidefinite: $\mathbf{x}^\top A \mathbf{x} \geq 0$ for all $\mathbf{x}$.
4. Negative semidefinite: $\mathbf{x}^\top A \mathbf{x} \leq 0$ for all $\mathbf{x}$.
5. Indefinite: takes both positive and negative values.

Example:

$$\mathbf{x}^\top \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \mathbf{x} = (x - y)^2 \geq 0$$

so the matrix is positive semidefinite.

Eigenvalue and minor tests

Spectral test (symmetric A):

- PD $\iff$ all eigenvalues > 0 ; ND $\iff$ all < 0 ;
- PSD $\iff$ all ≥ 0 ; NSD $\iff$ all ≤ 0 ;
- Indefinite $\iff$ eigenvalues of different signs.

Sylvester's criterion (leading principal minors D_k):

- PD $\iff D_k > 0$ for $k = 1, \dots, n$;
- ND $\iff (-1)^k D_k > 0$ for $k = 1, \dots, n$.

Example: $A = \begin{pmatrix} -3 & 2 & 0 \\ 2 & -3 & 0 \\ 0 & 0 & -5 \end{pmatrix}$ has $D_1 = -3$, $D_2 = 5$, $D_3 = -25 \Rightarrow$
negative definite.

Examples

1. $A = \begin{pmatrix} 1 & 4 & 6 \\ 4 & 2 & 1 \\ 6 & 1 & 6 \end{pmatrix}$: $D_1 = 1$, $D_2 = -14$, $D_3 = -109 \Rightarrow$ indefinite.

2. $A = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix}$: $D_1 = 3$, $D_2 = 5$, $D_3 = 12 \Rightarrow$ positive definite.

3. The symmetric matrix of $Q(x, y, z) = xy + yz$ is indefinite since $D_1 = D_3 = 0$ and $D_2 = -\frac{1}{4}$.